QUICK-REFERENCE GUIDE · FOR COACHES AND SPORTS BUSINESSES

Machine learning or deep learning?

Deep learning isn't automatically better. Answer four questions about your problem to pick the right approach before you spend time or money.

Question	MACHINE LEARNING e.g. SVM, decision trees	DEEP LEARNING e.g. CNNs, neural networks
1. What does your data look like?	Rows and columns: spreadsheets, stats, payments, attendance.	Raw images, video, audio, or text.
2. Can you name the features that matter?	Yes. You already track them (contract type, sessions attended).	No. They're too subtle to describe by hand (face shape, body mechanics).
3. Do you need to explain the "why"?	Yes. Simpler models show which factors drove a prediction.	Harder. Deep networks are often a "black box."
4. How much data and computing power?	Thousands of rows. Cheap and fast to retrain.	Large datasets and heavy computing (GPUs or built-in neural chips).

Rule of thumb: if your data fits in a spreadsheet, start with machine learning. Reach for deep learning when the data is raw and the patterns are too subtle to describe.

What it looks like in sports

PROBLEM	DATA	FIT	WHY
Which training clients won't renew their package	Sessions attended, package size, months active	ML	Same shape as telecom churn, and you need the reason to act on it.
Flagging players with high training load	Minutes, GPS load, effort ratings	ML	Structured numbers; coaches need to see which factor drove the flag.
Building a scouting shortlist from stats	Save %, distribution accuracy, minutes	ML	Known, meaningful columns. A simple model is fast and explainable.
Analyzing a goalkeeper's diving technique	Training or match video	DL	Body positions change every frame, the same reason Face ID needs a CNN.
Auto-cutting highlight clips from footage	Full match video	DL	The model has to "see" saves and goals in raw video.

97%
accuracy for an SVM predicting telecom churn on 7,043 customer records (Omari et al., 2025)

45 datasets
where tree-based models stayed state-of-the-art on medium-sized tables (Grinsztajn et al., 2022)

1 in 1,000,000
odds a stranger unlocks Face ID, powered by neural networks (Apple, 2026)

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Sports examples are my applied analysis, not tested models.
Full paper and sources on the portfolio page.